Groups Are Factors: Learning Linear PDEs by Clustering Modes on the Factors of the Symbol

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Abstract. Methods that learn partial differential equations from data usually regress a time derivative on a library of terms. We argue that for linear equations with constant coefficients the natural unit is not the term but the factor of the symbol. Every Fourier mode of a solution lies on the zero set of one factor, so learning the equation means grouping the modes by factor. We prove that every solution splits into one component per factor, uniquely up to an overlap counted by the intersection points of the zero sets; that additive terms admit no such split; that the split is stable exactly when the factors' solution spaces are well conditioned (for two factors, when their Friedrichs angle is positive); and that finite data can support but never certify an equation. The resulting algorithm groups twice: within one sample of the field it clusters Fourier modes onto the curves of the factors, and across posterior samples it clusters the factors with a Dirichlet-process mixture. Its output is a distribution over equations with intervals, the separation of each group, and a label stating what the data support. On 285 random equations never used in development it returns the right or an equivalent equation on 96.5% and is never wrong without warning, while weak SINDy is right on 28.4% and wrong without warning on 204. On a ten-equation development benchmark it recovers all thirty factor sets, including the products of factors on which the best library regression fails.

Key words. equation discovery, operator factorization, dispersion relation, Friedrichs angle, matrix pencil, Dirichlet process, uncertainty quantification

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1. Introduction.

Consider a quantity that is carried by a flow and diffuses at the same time, observed with noise. A natural model is

$$(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)\,u = 0, \tag{1.1}$$

whose solutions are sums of a travelling profile and a diffusing one. A method that regresses u_t on a library of terms such as u_x, u_{xx}, u_{xxx} looks for one equation $u_t = \sum_k \beta_k\,\partial_x^k u$. The data obey the *product* of two such equations, and no single sum of terms fits them: every library-regression method we tested returns a wrong equation for (1.1) at 5% noise, and none of them warns.

The idea. In Fourier variables a constant-coefficient operator is a polynomial in (τ, ξ), with $\tau \leftrightarrow \partial_t$ and $\xi \leftrightarrow \partial_x$; the operator in (1.1) is $(\tau + c\,\xi)(\tau - \nu\,\xi^2)$. A Fourier mode $e^{\lambda t + i\kappa x}$ solves the equation exactly when $(\lambda, i\kappa)$ is a zero of this polynomial, that is, a zero of one of its factors. So the modes of every solution lie on a few curves, one per factor: here the line $\lambda = -ic\kappa$ (transport) and the parabola $\lambda = -\nu\kappa^2$ (diffusion), as in Figure 1. *Learning the equation means finding these curves, which is a clustering problem whose clusters are curves rather than points.* This paper develops that idea from the algebra to an algorithm with calibrated uncertainty.

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(a) oscillation (b) damping

Figure 1. *The idea of the paper on* $(\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)u = 0$. *Every Fourier mode* $e^{\lambda t + i\kappa x}$ *of a solution lies on the zero set of one factor of the symbol* $(\tau + c\xi)(\tau - \nu\xi^2)$*: circles on the line of the transport factor, squares on the parabola of the diffusion factor (schematic; Figure 4 shows the modes found in noisy data). Learning the equation means learning these two groups.*

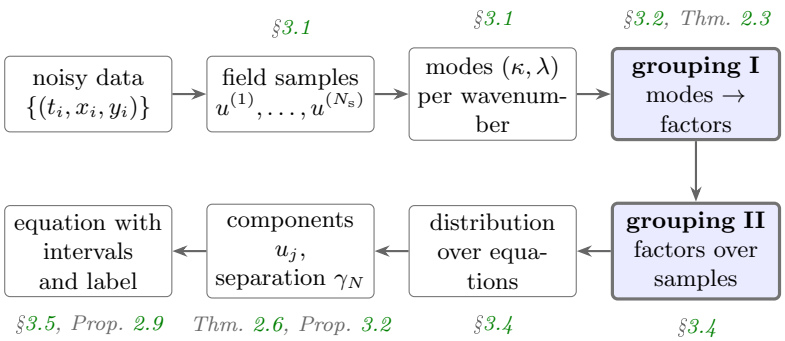


Figure 2. *The method at a glance. The two grouping steps (thick boxes) are the core of the paper: modes are clustered onto the curves of the factors within one sample, and factors are clustered across posterior samples. Each box names the section that describes it and the result that justifies it.*

What the theory says. Three questions decide whether this clustering is well posed, and section 2 answers each. *Are the groups really the factors?* Yes: every solution splits into one component per factor, uniquely up to an overlap whose dimension is the number of intersection points of the zero sets (Theorem 2.3); additive terms, by contrast, admit no such split (Proposition 2.5). *How well can the groups be separated?* Exactly as well as the Friedrichs angle between the factors' solution spaces allows (Theorem 2.6). *What can finite data establish?* They can support a factored equation but never certify it (Proposition 2.9); this is why the algorithm reports labels, not "identified".

What the algorithm does. It groups twice (Figure 2). *Within one sample* of the field it reads the temporal rates of every wavenumber (the points of Figure 1) and clusters them onto the curves of candidate factors. *Across posterior samples* it clusters the factors found in each sample, with a Dirichlet-process mixture, so that the uncertainty of the data becomes a distribution over equations. Every decision is finally checked against the raw observations, and the output carries intervals, the separation of each group and a label.

Contributions.

1. *The grouping principle and its limits*: groups are factors, with the exact ambiguity $\delta(P)$ (Theorem 2.3); terms are not groups (Proposition 2.5); the conditioning of the split (Theorem 2.6) and its Bayesian form (Proposition 3.2); what finite data can certify (Theorem 2.8 and Proposition 2.9).
2. *An algorithm that learns the groups* from noisy, gridded or scattered data, by clustering modes onto curves and factors across samples, whose output encodes these limits as intervals and labels.
3. *Evidence*: on 285 random equations never used in development, 96.5% right or equivalent and no confident wrong answer, against 28.4% and 204 confident wrong answers for weak SINDy; on a ten-equation benchmark in three designs, 30 of 30 factor sets.

Relation to prior work. Most methods that learn a PDE from data regress a time derivative on a library of candidate terms: SINDy and PDE-FIND [2, 16], their weak, ensemble and Bayesian variants [11, 5, 7], and Gaussian-process formulations of linear operators [15]. Their unit is the *term*. Dynamic mode decomposition [18, 17] works with eigenvalues, as we do, but returns a linear operator on the data, not an equation, and does not group eigenvalues by the factor they belong to. Matrix pencils [9], the algebra of constant-coefficient operators and the Ehrenpreis–Palamodov fundamental principle [4, 13, 10, 8, 12], and Dirichlet-process mixtures [1] are classical. What is new is the connection: the *factor* of the symbol, not the term, as the unit of learning; what finite data can certify about it and how stably it can be separated; and an algorithm that groups twice and reports these limits.

Scope. Everything proved and implemented concerns scalar linear equations with constant coefficients in one time and one space variable, observed on a rectangle. Equation (1.1) with $c = 1$, $\nu = 0.3$ is the running example. The proofs of the main results, the complete algorithm with its parameters, further cases, a guide to reading the output, and the code and benchmark data are in the supplementary materials; the code is public [14].

2. The grouping principle.

This section makes the picture of Figure 1 precise. Operators become polynomials (subsection 2.1); the factors of the polynomial split every solution, and terms do not (subsection 2.2); the split is as stable as the angle between the factors' solution spaces (subsection 2.3); and finite data can support but not certify the result (subsection 2.4). Each subsection ends with the rule it gives to the algorithm. Proofs are in section SM2.

2.1. Derivatives become polynomials.

Every operator we consider has constant coefficients, $P(\partial_t, \partial_x) = \sum_{j,k} c_{jk}\, \partial_t^j \partial_x^k$ with $c_{jk} \in \mathbb{C}$. The exponentials $\psi_{\tau,\xi}(t,x) = e^{\tau t + \xi x}$ are common eigenfunctions of all such operators:

$$P(\partial)\, \psi_{\tau,\xi} = P(\tau,\xi)\, \psi_{\tau,\xi}. \tag{2.1}$$

Differentiation becomes multiplication, and operators become polynomials.

Definition 2.1 (Symbol and characteristic variety). *The constant-coefficient operators form the commutative ring* $\mathcal{R} = \mathbb{C}[\partial_t, \partial_x] \cong \mathbb{C}[\tau, \xi]$. *The polynomial* $P(\tau,\xi)$ *is the* symbol *of* $P(\partial)$, *and its zero set* $V(P) = \{(\tau,\xi) \in \mathbb{C}^2 : P(\tau,\xi) = 0\}$ *is the* characteristic variety.

By (2.1), $\psi_{\tau,\xi}$ solves $P(\partial)u = 0$ exactly when $(\tau,\xi) \in V(P)$. A Fourier mode in space is the case $\xi = i\kappa$ with wavenumber $\kappa \in \mathbb{R}$; then $P(\lambda, i\kappa) = 0$ is the *dispersion relation*, and its

roots $\lambda(\kappa)$ are the *temporal rates* of the modes $e^{\lambda(\kappa)t+i\kappa x}$. The real part of a rate is growth or decay, the imaginary part is oscillation. We keep ξ in statements about the ring and set $\xi = i\kappa$ only on Fourier modes, so that a real equation has a symbol with real coefficients.

Example 2.2 (Running example). $P = (\partial_t + c\,\partial_x)(\partial_t - \nu\,\partial_x^2)$ has symbol $P = Q_1Q_2$ with $Q_1 = \tau + c\,\xi$ and $Q_2 = \tau - \nu\,\xi^2$. On Fourier modes the two factors give two branches of rates, $\lambda_1(\kappa) = -ic\kappa$ (a wave that travels without decay) and $\lambda_2(\kappa) = -\nu\kappa^2$ (a standing mode that decays). A solution is any superposition of modes on the two branches. The equation is one object; its two factors are two groups of modes.

2.2. Groups are factors, not terms. Let the symbol be factored into pairwise coprime factors,

$$P = Q_1 \cdots Q_N, \qquad Q_j = P_j^{e_j}, \tag{2.2}$$

with P_j irreducible and e_j its multiplicity. Two ways of splitting a solution suggest themselves: one piece per factor Q_j, or one piece per additive term of the operator (for $u_t = \nu u_{xx} - cu_x$, one piece for diffusion and one for transport). Only the first works.

Theorem 2.3 (Factor decomposition). *Let $\Omega \subset \mathbb{R}^2$ be bounded, connected and simply connected, and let P be as in (2.2). Every solution u of $P(\partial)u = 0$ on Ω (a distribution or a smooth function) can be written*

$$u = \sum_{j=1}^{N} u_j, \qquad Q_j(\partial)\, u_j = 0. \tag{2.3}$$

The tuples $(u_1, \dots, u_N)$ that sum to zero form a space of exponential-polynomials of dimension exactly

$$\delta(P) = \sum_{i<j} \dim_{\mathbb{C}} \mathcal{R}/(Q_i, Q_j), \tag{2.4}$$

the number of intersection points of the varieties $V(Q_i)$, counted pair by pair and with multiplicity.

The theorem says that the decomposition *exists* and that its only freedom has finite dimension. Each meeting point of two curves $V(Q_i)$ and $V(Q_j)$ is a mode that solves both factors and can therefore be moved from one group to the other without changing u. In one variable the statement is Bézout's identity: from $aQ_1 + bQ_2 = 1$ follows the split $u = bQ_2u + aQ_1u$. In two variables coprime factors no longer generate 1; they only have finitely many common zeros, and $\delta(P)$ counts exactly that defect. The hypotheses matter: on a domain with a hole the Laplacian factors as $(\partial_x + i\partial_y)(\partial_x - i\partial_y)$, yet its solution $\log|z|^2$ on an annulus is not a sum of a holomorphic and an antiholomorphic single-valued function. This is why the algorithm works on rectangles.

Example 2.4 (Running example: two groups, two shared modes). $Q_1 = \tau + c\xi$ and $Q_2 = \tau - \nu\xi^2$ meet at $(0, 0)$ and $(c^2/\nu, -c/\nu)$, so $\delta(P) = 2$: a constant and the exponential $e^{(c^2/\nu)t-(c/\nu)x}$ can be moved between the groups. Among Fourier modes only the first is visible: the $\kappa = 0$ mode belongs to both groups.

The groups are read from the rates. When every factor is an evolution factor, $Q_j = \tau - \Lambda_j(\xi)$, the component u_j is obtained from u by a Fourier multiplier. On a wavenumber κ at which no two branches meet,

$$\hat{u}_j(t,\kappa) = \prod_{k \neq j} \frac{\partial_t - \Lambda_k(i\kappa)}{\Lambda_j(i\kappa) - \Lambda_k(i\kappa)}\, \hat{u}(t,\kappa), \tag{2.5}$$

the Lagrange formula for the spectral projectors of ∂_t restricted to that wavenumber. *The rates decide the groups*, and the poles of the multiplier sit exactly at the visible part of the overlap $\delta(P)$. This is what the algorithm exploits: it never needs the operator in advance, only the rates and their arrangement on curves.

Why additive terms do not form groups.

Proposition 2.5 (Terms do not give components). *If $P = A_1 + A_2$ with constant coefficients, then every solution of $P(\partial)u = 0$ that is a sum $a + b$ with $A_1 a = 0$ and $A_2 b = 0$ also solves $A_1 A_2 u = 0$. If $V(A_1) \cap V(A_2)$ is finite, these solutions form a finite-dimensional space of exponential-polynomials.*

Proof sketch. $A_1A_2(a+b) = A_2A_1a + A_1A_2b = 0$ because the operators commute. A common zero of P and A_1A_2 is a common zero of A_1 and A_2, so the joint kernel of P and A_1A_2 lives on a finite set of exponents. ∎

For the heat equation, $A_1 = \partial_t$ and $A_2 = -\nu\partial_x^2$: a "stationary part plus a diffusion part" $f(x) + a(t) + b(t)x$ is killed by $\partial_t\partial_x^2$, while a single heat mode $e^{-\nu\kappa^2 t}\cos\kappa x$ is not. A generic solution is not a sum of single-term pieces. What a term-wise split leaves over is the *interaction* of the terms; it is as large as the solution itself, and on a non-periodic window it cannot be removed by any continuous, hence any regularized, procedure (section SM3). Different factors, in contrast, superpose exactly: *factors, not terms, are the groups.*

Role in the algorithm. The objects to cluster are modes, and a cluster is a factor: a curve $V(Q_j)$ in the plane of rates. The overlap $\delta(P)$ is the irreducible ambiguity; the algorithm resolves it by a fixed convention (subsection 3.2). Because the interaction of terms cannot be regularized away, the algorithm uses no coefficient penalty.

2.3. How well separated are the groups? The Friedrichs angle. Theorem 2.3 says that the split exists, not that it is stable. Let $M_j = \ker_{H^s} Q_j$ be the solutions of factor j in the Sobolev space $H^s(\Omega)$, let $\mathcal{S}(u_1, \ldots, u_N) = \sum_j u_j$ be the sum operator, and define its conditioning constant

$$\gamma_N = \inf_{0 \neq v \perp \ker \mathcal{S}} \frac{\|\mathcal{S}v\|^2}{\|v\|^2}, \qquad \|(u_1,\ldots,u_N)\|^2 = \sum_j \|u_j\|_{H^s}^2. \tag{2.6}$$

It is the worst case, over all tuples of pieces, of the size of the sum relative to the size of the pieces. If the spaces are orthogonal, $\gamma_N = 1$; if M_2 approaches M_1, one can take $u_2 \approx -u_1$, the pieces stay of unit size while their sum vanishes, and $\gamma_N \to 0$. For two groups $\gamma_2 = 1 - c_F$, where $c_F = \cos\theta_F$ is the cosine of the smallest nonzero angle between M_1 and M_2, the *Friedrichs angle* [3] (Figure 3). In regression terms M_1 and M_2 are two blocks of regressors, and c_F is the largest correlation between a combination of the first block and one of the second.

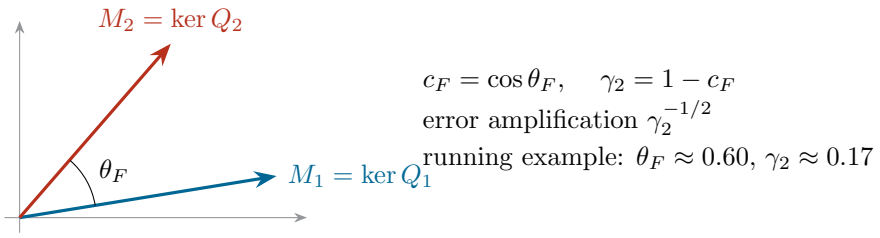


Figure 3. *The Friedrichs angle between the solution spaces of two factors. A small angle means that large pieces can almost cancel, so recovering the pieces from their sum amplifies errors by up to* $\gamma_2^{-1/2} = (1 - \cos\theta_F)^{-1/2}$. *The values are those measured on the running example (subsection 4.1).*

Theorem 2.6 (Stability of the split). *Every* $u \in \ker_{H^s} P$ *has components* $u_j \in H^s$ *if and only if* $\sum_j M_j$ *is closed and dense in* $\ker_{H^s} P$. *Closedness is equivalent to* $\gamma_N > 0$, *and then the minimum-norm components satisfy*

$$\Big(\sum_j \|u_j\|_{H^s}^2\Big)^{1/2} \le \gamma_N^{-1/2}\, \|u\|_{H^s}. \tag{2.7}$$

For the finite-dimensional sampled kernels used by the algorithm the sum is always closed, and stability reduces to $\gamma_N > 0$.

Two remarks keep the theorem honest. On infinite-dimensional component spaces $\gamma_N > 0$ can fail, and the pieces of a smooth u may be one derivative rougher than u. For $N \ge 3$ groups the pairwise angles do not determine γ_N: three spaces can be pairwise well separated and jointly nearly dependent, so the full Gram operator of $\mathcal{S}$ is needed.

Role in the algorithm. γ_N is computed on the sampled kernels and reported with every result. It sets the width of the intervals of the components (Proposition 3.2): closely aligned groups are never hidden; they show up as wide intervals, not as a confident but wrong split.

2.4. What finite data can and cannot certify.

When does a solution determine its equation? Let the *annihilator* $\mathrm{Ann}(u) = \{Q \in \mathcal{R} : Q(\partial)u = 0\}$ collect every constant-coefficient equation that u satisfies; it is an ideal of $\mathcal{R}$. The question is whether $\mathrm{Ann}(u) = (P)$, i.e. whether every equation u satisfies is a consequence of P. The answer turns on one class of functions.

Definition 2.7 (Exponential-polynomials). EP *is the set of finite sums* $\sum_j p_j(t,x)\, e^{\tau_j t + \xi_j x}$ *with polynomials* p_j *and exponents* $(\tau_j, \xi_j) \in \mathbb{C}^2$. *Equivalently (section SM2),* $w \in \mathrm{EP}$ *if and only if the derivatives of* w *span a finite-dimensional space.*

An exponential-polynomial *remembers too little*: it sits on finitely many points of $\mathbb{C}^2$, and through finitely many points pass infinitely many curves, so infinitely many equations. A single heat mode $e^{-\nu\kappa^2 t + i\kappa x}$ solves the heat equation, but equally the pure decay $\partial_t + \nu\kappa^2$ and every other equation whose variety passes through $(-\nu\kappa^2, i\kappa)$. The heat kernel $t^{-1/2} e^{-x^2/(4\nu t)}$, whose derivatives span an infinite-dimensional space, fills the whole curve, and every equation it satisfies is a multiple of $\partial_t - \nu\partial_x^2$. The general statement strips one irreducible factor at a time.

Theorem 2.8 (Identification). *Let Ω be open and connected, $P(\partial)u = 0$, and $P = \prod_i P_i^{e_i}$ with distinct irreducible P_i. Then $\mathrm{Ann}(u) = (P)$ if and only if $(P/P_i)(\partial)u \notin \mathrm{EP}(\Omega)$ for every i. In that case P is the unique annihilator of minimal degree, up to a constant.*

In words: u determines its equation exactly when, after removing any one factor, what remains is still a genuine solution of that factor, not a finite sum of modes. Data, however, are finitely many numbers, and finitely many numbers are always matched by an exponential-polynomial.

Proposition 2.9 (Finite data). *Let Ω be connected and $z_0 \in \Omega$. (i) If the derivatives of u span a space of dimension at most n, the derivatives of order at most $2n-1$ at z_0 determine $\mathrm{Ann}(u)$. (ii) Without such a bound, for every finite set of derivatives at a point, and for every finite set of samples, there is an exponential-polynomial that matches them; hence no finite data prove $\mathrm{Ann}(u) = (P)$.*

Role in the algorithm. An algorithm can refute a candidate equation, or report that it is compatible with the data at the available resolution, but it can never certify it. The output therefore carries a label (subsection 3.5), and the label that plays the role of "identified" requires each factor to be seen at enough distinct wavenumbers to pin it down within its class (Proposition 3.1).

3. Learning the groups.

The algorithm turns the data into modes (subsection 3.1), clusters the modes onto the curves of factors within each sample of the field (grouping I, subsection 3.2) inside a class of operators chosen by the data (subsection 3.3), clusters the factors across samples (grouping II, subsection 3.4), and labels the result (subsection 3.5). The complete algorithm, with every parameter and its default, is in section SM4.

3.1. From data to modes.

Field samples. The input is a set of noisy observations $y_i = u(t_i, x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, on a window $[0, T] \times [0, L)$ periodic in x, or a noisy field on a regular grid. Scattered observations are interpolated by a Gaussian process whose kernel is a sum of separable Matérn-5/2 products (a transported profile and a diffusing one do not share a length scale), with the hyperparameters integrated out by importance sampling. The output of this stage is not the posterior mean but N_s posterior *samples* $u^{(q)}$ of the field, each with a weight π_q. The reason is structural: the maps applied next are not continuous (factors of a numerical symbol jump where it becomes reducible), so the mean of the outputs differs from the output of the mean, and the mean, a Wiener filter, shrinks exactly the fast-decaying high-wavenumber modes that carry the diffusive factors. On gridded input the noisy grid itself is used, and parametric-bootstrap refits provide the spread.

The usable band. Not every wavenumber carries information about the symbol. Three limits apply: spatial aliasing ($|\kappa| \le \pi/\Delta x$); temporal aliasing along a branch, since a rate is read from samples spaced by Δt only modulo $2\pi i/\Delta t$; and, usually the binding one, noise together with decay. After the spatial Fourier transform one wavenumber bin carries noise of standard deviation $\sigma_\kappa = \sigma_n/\sqrt{n_x}$, and a mode is used only while its amplitude exceeds $z\,\sigma_\kappa$ ($z = 3$). A diffusive mode of rate $-\nu\kappa^2$ falls below this threshold after a time of order $\log(|A_\kappa|/z\sigma_\kappa)/(\nu\kappa^2)$, so the high wavenumbers of a dissipative group are lost first. This is a property of the data, not of the method. The band $\mathcal{K}$ is the set of wavenumbers that pass all

three limits. It must hold enough distinct wavenumbers per group:

Proposition 3.1 (Counting condition). *Let $Q = \tau - \Lambda(\xi)$ with Λ a polynomial of degree at most d. Its coefficients are determined by the rates $\lambda(\kappa_\ell) = \Lambda(i\kappa_\ell)$ at $d+1$ distinct wavenumbers, and not by fewer; the coefficient error is at most $\|V^{-1}\| \max_\ell |\delta\lambda_\ell|$ with $V = ((i\kappa_\ell)^k)$ the Vandermonde matrix, whose inverse grows like h_κ^{-d} when the nodes cluster within a spread h_κ [6]. For real coefficients $\lfloor d/2 \rfloor + 1$ distinct values of $|\kappa|$ suffice.*

So the whole band is used, not the cleanest few bins: spreading the wavenumbers is what keeps V well conditioned.

Rates by the matrix pencil. For each sample and each $\kappa \in \mathcal{K}$, the spatial Fourier transform gives a time series that, for a constant-coefficient equation, is a sum of exponentials, one per root of the symbol at that wavenumber:

$$\hat{u}(t,\kappa) = \sum_{p=1}^{r(\kappa)} A_p \, e^{\lambda_p(\kappa) t}, \qquad Q_{j(p)}\big(\lambda_p, i\kappa\big) = 0. \tag{3.1}$$

The rates are the eigenvalues of ∂_t on the space spanned by the series and its shifts, and the matrix pencil method computes them [9]. From the Hankel matrix $H_{\ell k} = \hat{u}(t_{\ell+k}, \kappa)$ take Y_0 (H without its last column) and Y_1 (H without its first), both projected on the r leading singular directions. Advancing a series by one step multiplies the term $A_p e^{\lambda_p t}$ by $\mu_p = e^{\lambda_p \Delta t}$, so

$$\lambda_p = \tfrac{1}{\Delta t} \log \mu_p, \qquad \mu_p \in \operatorname{eig}\big(Y_0^{+} Y_1\big). \tag{3.2}$$

The order r is chosen per bin by BIC, and each rate receives a standard deviation $\sigma_{\lambda,p}$ from the Cramér–Rao bound of model (3.1); weak, fast-decaying or nearly coincident modes receive large ones. This step does *not* fit the equation: it returns a cloud of points $(\kappa, \lambda_p(\kappa))$ with error bars, the noisy version of Figure 1. Two rates closer than about $4/T$ cannot be resolved in a window of length T; such bins are flagged as confluent.

3.2. Grouping I: modes onto curves. The obvious route, fitting one polynomial through all points and factoring it, is ill-posed: a polynomial with noisy coefficients is almost never reducible, and when it is, its factors move discontinuously with the coefficients (in our tests this route split the irreducible Klein–Gordon and telegraph symbols at noise 10^{-4}). We therefore group and fit at the same time. Given a structure M (the number N of factors and their degrees, subsection 3.3), we solve the *N-varieties problem*, a version of k-means in which the cluster centres are curves instead of points: assign every mode to the factor whose variety is nearest, refit each factor to the modes assigned to it, and repeat until the assignment stops changing. The distance of mode p to factor Q_j is its first-order distance to the variety in units of its own uncertainty,

$$r_j(p) = \frac{|Q_j(\lambda_p, i\kappa_p)|}{\sigma_{\lambda,p}\, |\partial_\tau Q_j(\lambda_p, i\kappa_p)|}, \tag{3.3}$$

and each mode costs $\min(r_j(p)^2, 9)$, so an isolated outlier pays a fixed penalty instead of dragging a curve away. This is the grouping of Theorem 2.3 learned directly from the spectrum.

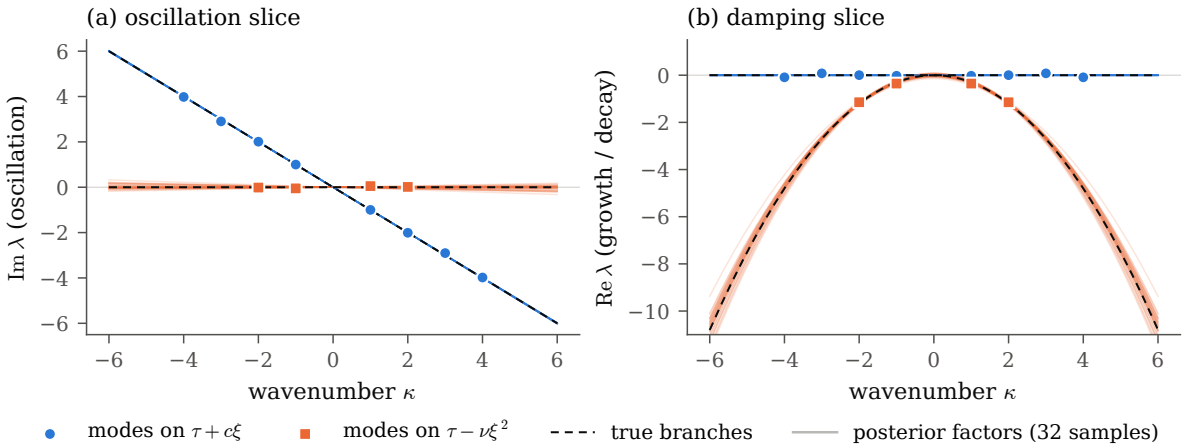


Figure 4. *Grouping I on the running example (64 × 128 grid, 5% noise). Points: modes found by the matrix pencil, coloured by the factor they were assigned to. Dashed: the true branches $\lambda = -ic\kappa$ and $\lambda = -\nu\kappa^2$. Thin lines: the factors fitted to 32 posterior samples; they bundle where there are modes and fan out where there are none. (a) Oscillation* $\operatorname{Im}\lambda$*: transport is the sloped line, diffusion the flat one. (b) Damping* $\operatorname{Re}\lambda$*: transport is flat at zero, diffusion the downward parabola.*

Figure 4 shows the result on the running example. Each factor is a direction in the plane of rates, and the modes of one group line up along it. Once modes are assigned, the component of group j is the sum of its modes, $u_j = \sum_{p:j(p)=j} A_p\, e^{\lambda_p(t-t_0)+i\kappa_p x}$, the discrete counterpart of the projector (2.5). A mode within distance 3 of two varieties lies in the overlap $\delta(P)$ (for the running example, $\kappa = 0$) and is split equally between the groups: among all splits allowed by Theorem 2.3 this one puts the least energy into the ambiguity, and it coincides with the posterior mean of Proposition 3.2. The energy shares $w_j = \|u_j\| / \sum_k \|u_k\|$ follow. A group can itself split further (two conjugate branches of the beam equation are first found as one factor of degree two in τ); each component is therefore analysed again, and a split is kept only if it leaves the kernel unchanged or improves the evidence of subsection 3.5.

3.3. Choosing the class of operators. A structure $\mathsf{M} = ((m_1, d_1), \dots, (m_N, d_N))$ fixes N and, for each factor, its degree m_j in τ and d_j in ξ; the running example is $((1,1),(1,2))$. The class $C(m,d)$ holds all structures with $\sum_j m_j \le m$ and $d_j \le d$. Grouping I is run for every structure, and structures are compared by

(3.4)
$$\mathrm{BIC}(\mathsf{M}) = 2n_m \log \max\!\big(\chi^2/(2n_m),\, 1\big) + n_{\mathrm{par}}(\mathsf{M}) \log(2n_m), \qquad P(\mathsf{M} \mid \text{data}) \propto e^{-\mathrm{BIC}(\mathsf{M})/2},$$

where χ^2 is the total capped misfit of the n_m modes and n_{par} the number of coefficients. The floor $\max(\cdot, 1)$ means that misfit is penalized only once it exceeds what the noise explains, so a fit better than the noise is never rewarded. The class is grown from $C(3,4)$ (90 structures) up to $C(4,8)$ (1353) only when the data demand it, i.e. when most of the posterior mass is on structures that fail the tests of subsection 3.5, and a larger class is kept only if it also predicts the second half of the time window from the first. This is the only regularization of the method, and it acts on the class, by Occam's razor, not on the coefficients: by subsection 2.2

a coefficient penalty cannot remove the interaction of terms, and by Theorem 2.6 it can push the error of the split below its floor only by shrinking the components, which the next layer does explicitly and reports.

3.4. Grouping II: factors across samples. Grouping I gives one set of factors per field sample. Two questions remain: how well are the components determined given the factors, and which factors of different samples are the same factor?

The angle sets the error bars. On the observation window write $u = \sum_j B_j a_j + \varepsilon = \mathcal{S}a + \varepsilon$, with B_j an orthonormal basis of the sampled kernel of Q_j, $\varepsilon \sim \mathcal{N}(0, \sigma_n^2 I)$ and a Gaussian prior $a \sim \mathcal{N}(0, \omega^2 I)$, whose scale ω^2 is set by empirical Bayes. The posterior of the coefficients is Gaussian, $\mathcal{N}(\hat{a}, \Gamma_a)$ with $\Gamma_a = \sigma_n^2(\mathcal{S}^H\mathcal{S} + (\sigma_n^2/\omega^2)I)^{-1}$; the layer returns a distribution of splits, not one split.

Proposition 3.2 (The angle sets the error bars). *For two groups the eigenvalues of $\mathcal{S}^H\mathcal{S}$ are $1 \pm \cos\theta_k$, with θ_k the principal angles between the sampled kernels (and 1 on the directions of the larger kernel without a partner). In the worst direction orthogonal to $\ker \mathcal{S}$ the posterior variance of the component coefficients is*

$$\frac{\sigma_n^2}{\gamma_2 + \sigma_n^2/\omega^2}, \qquad \gamma_2 = 1 - c_F. \tag{3.5}$$

On the shared directions ($\theta_k = 0$, the overlap $\delta(P)$) the posterior equals the prior, and its mean splits the shared part equally.

This is the Bayesian form of (2.7): intervals grow like $\sigma_n/\sin(\theta_F/2)$ as the angle closes and are capped by the prior when $\theta_F \to 0$. The same evidence compares the two-factor model with the one-factor model and gives the probability $P(\text{split})$, which returns to its prior as $\theta_F \to 0$.

The cloud of factors. Each factor of each sample is mapped to the vector β of its monic coefficients, zero-padded to the largest degrees in play, and receives the mass of its sample times its energy share:

$$W = \sum_{q=1}^{N_s} \pi_q \sum_j w_j^{(q)}\, \delta_{\beta_j^{(q)}}. \tag{3.6}$$

A sharp equation is a cloud with a few tight clusters, an uncertain one a diffuse cloud. Since the number of groups is unknown, the cloud is modelled by a Dirichlet-process Gaussian mixture [1] (truncated stick-breaking, variational inference), which lets the data choose the number of clusters; two clusters that never occur in the same sample are merged when one Gaussian explains them better. The cluster centres are the factors, quantiles over samples give their 90% intervals, and the cluster responsibilities give the posterior $P(k)$ of the number k of groups. Figure 5 shows the cloud of the running example.

3.5. Decision and labels. The interpolated field has already smoothed the high wavenumbers, so the final decisions are made on the *raw observations*. A candidate set of factors defines a linear model $y = Ea + \varepsilon$ for the n observations, where the columns of E are the kernel exponentials of the factors evaluated at the observation points. Integrating the amplitudes out

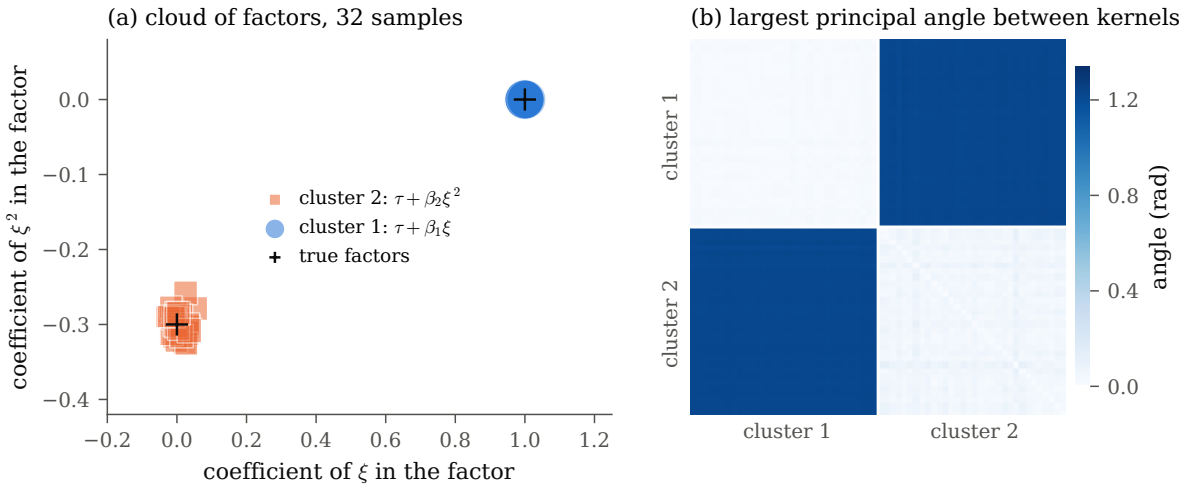


Figure 5. *Grouping II on the running example. (a) Each marker is one factor found in one of* 32 *posterior samples, placed by its coefficients of ξ and ξ^2; area is proportional to its mass $\pi_q w_j$; crosses mark the true factors. The cloud has two tight clusters and nothing in between. (b) Largest principal angle between the sampled kernels of every pair of factors, ordered by cluster: factors of the same group allow the same functions (light diagonal blocks), factors of different groups do not (dark off-diagonal blocks).*

gives $y \mid Q_1, \dots, Q_N \sim \mathcal{N}(0, \sigma_n^2 I + \omega^2 E E^H)$, and the score

$$S(Q_1, \dots, Q_N) = \log \mathcal{N}\big(y; 0, \sigma_n^2 I + \omega^2 E E^H\big) - \tfrac{1}{2} n_{\text{par}} \log n \tag{3.7}$$

decides between dropping a factor, dropping a term whose interval contains zero, merging two factors or splitting one; a change needs more than one nat. The same model gives a goodness-of-fit statistic $\chi^2 = \sum_i |y_i - \hat{y}_i|^2 / \sigma_n^2$, approximately chi-squared with $n - \operatorname{rank} E$ degrees of freedom. The result then receives one of four labels, which encode what subsection 2.4 allows:

FIT-COMPATIBLE	every factor has more modes than coefficients, at least $d_j + 1$ distinct wavenumbers (Proposition 3.1), and the fit passes the goodness-of-fit test: the data are compatible with the factored equation and pin it down within the class.
ID-KER	the fit is good, but some factor is pinned down only relative to the declared class.
MISFIT	the goodness-of-fit test rejects the structure ($p < 10^{-3}$).
UNDECIDED	the data cannot decide, typically because of confluent rates or a repeated factor P_i^2, whose polynomial prefactors $t\, e^{\lambda t}$ finite noisy data cannot separate from two nearby rates.

The output is therefore a distribution, never a point estimate: the factors with 90% intervals, $P(k)$ and the posterior over structures, the energy shares w_j with intervals, $P(\text{split})$, the separation γ_N, and the label. A small Friedrichs angle does not reject the equation; it widens the intervals, so the reader always sees how well each group is separated, not only whether the equation fits.

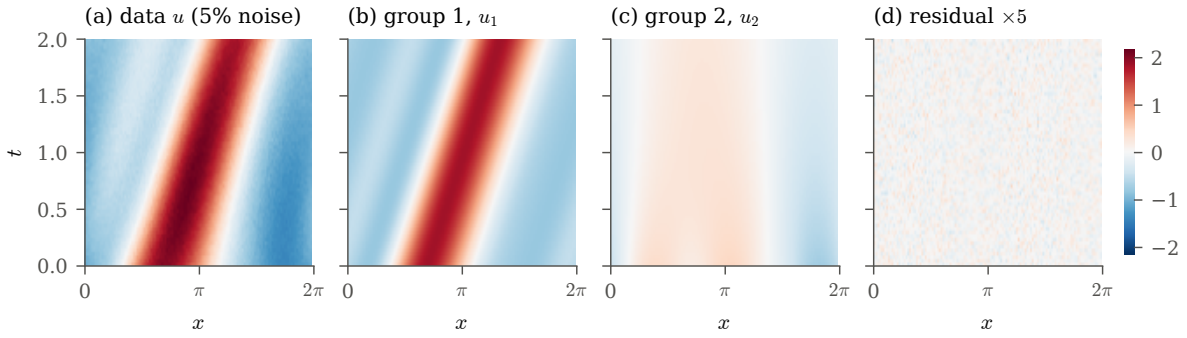


Figure 6. *The worked example as functions of (t, x) on a common colour scale: (a) noisy data; (b) transport component; (c) diffusion component; (d) residual, magnified five times, indistinguishable from the noise.*

4. Evidence. We first follow one data set through both groupings (subsection 4.1), then report the large benchmark on random equations and the development benchmark (subsection 4.2). An answer is *right* if the recovered set of factors matches the true one within relative coefficient error 0.05; *equivalent* if it explains the data as well as the true equation (a χ^2 test adjusted for degrees of freedom, or the same product written differently); an *abstention* if the label is ID-KER or UNDECIDED; and *false-confident* if a wrong equation is returned as if it were right. Library-regression baselines cannot abstain, so each of their wrong answers is false-confident.

4.1. One run, step by step. The data are the running example with $c = 1$, $\nu = 0.3$ on a 64×128 grid over $[0, 2] \times [0, 2\pi)$, with white noise of standard deviation $0.05\,\mathrm{std}|u|$. *Modes*: only $\kappa \leq 4$ rise above the detection threshold; the matrix pencil returns twelve modes. *Grouping I* (Figure 4): eight modes lie on the line $\mathrm{Im}\,\lambda = -\kappa$ with $\mathrm{Re}\,\lambda \approx 0$, four on the parabola $\mathrm{Re}\,\lambda = -0.3\kappa^2$; the transport modes occupy eight distinct wavenumbers and the diffusion modes four, against two and three required by Proposition 3.1. *Class*: the structure $((1, 1), (1, 2))$ ranks first. *Grouping II* (Figure 5): over 32 posterior samples the 64 factors form two tight clusters, with masses 0.767 and 0.233 and $P(k = 2) = 1$. *Components* (Figure 6): the transport component consists of tilted stripes, a profile moving at speed 1; the diffusion component has vertical structure fading in time; the residual, 2.5% of the data norm, shows no structure. Table 1 lists the output. The groups are only moderately separated ($\theta_F = 0.60$, amplification $\gamma_N^{-1/2} = 2.41$), and the intervals still cover the truth; the two conjugate factors of the beam equation, much closer to each other ($c_F = 0.939$), are recovered with the same accuracy and an amplification of 4.05 (section SM5).

4.2. Benchmarks.

Large benchmark on random equations. To test generalization we drew 300 equations at random from five families, 60 each: first-order factors (F1); first-order dispersive factors (F2); irreducible second-order factors (F3); products of two or three factors (F4); and third-order factors or a reaction factor times a transport factor, often with few excited wavenumbers (F5). Every draw is a different equation with an independent field and independent noise (1%

Table 1

Output of the worked example and how to read it. Intervals are 90% *intervals from* 16 *parametric-bootstrap refits.*

Quantity	Value	Meaning
equation	$(\tau + 1.0006\,\xi)(\tau - 0.2980\,\xi^2)$	recovered factors; truth $(\tau + \xi)(\tau - 0.3\,\xi^2)$
coefficient of ξ	1.0006 $[0.9998, 1.0020]$	speed c of the transport group
coefficient of ξ^2	-0.2980 $[-0.3049, -0.2919]$	diffusivity ν of the diffusion group
label	FIT-COMPATIBLE	enough distinct wavenumbers per factor; goodness of fit passed
$P(k)$; $P(\text{split})$	$P(k=2)=1$; 1.000	number of groups; evidence that both groups are needed
$w_1,\ w_2$	0.768, 0.232	energy shares of the two groups
$\gamma_N,\ \theta_F;\ \gamma_N^{-1/2}$	0.173, 0.60; 2.41	separation of the groups; worst-case error amplification

Table 2

Large benchmark: 285 *random equations not used in development, grid,* 1% *noise. "Right+eq." is the share of right or equivalent answers with a Wilson* 95% *interval; "abstain" includes one correct abstention (F2) and one crash (F4).*

family	method	n	right	equiv.	abstain	false-conf.	right+eq.
all	this method	285	187	88	10	**0**	**96.5%** [93.7, 98.1]
	weak SINDy	285	77	4	0	204	28.4% [23.5, 33.9]
F1	this method	60	59	1	0	0	100%
	weak SINDy	60	42	0	0	18	70.0%
F2	this method	60	31	23	6	0	90.0%
	weak SINDy	60	9	0	0	51	15.0%
F3	this method	60	41	18	1	0	98.3%
	weak SINDy	60	26	2	0	32	46.7%
F4	this method	48	12	34	2	0	95.8%
	weak SINDy	48	0	1	0	47	2.1%
F5	this method	57	44	12	1	0	98.2%
	weak SINDy	57	0	1	0	56	1.8%

of std$|u|$, regular grid), and none was used to set any parameter: the pipeline ran with the defaults fixed during development. We compare with weak SINDy [11], the strongest library regression on the development benchmark below, run through PySINDy with a linear library of mixed derivatives and a sparsity threshold chosen by BIC. 285 draws finished; 15 (twelve in F4, three in F5) exceeded the time limit of three hours per draw and are reported separately. Table 2 and Figure 7(a) give the outcomes.

The method is right or equivalent on 275 of 285 draws (96.5%) and false-confident on none; the exact one-sided 95% upper bound on its false-confident rate is 1.0%. Weak SINDy is right or equivalent on 81 (28.4%) and false-confident on 204. On the same draws the method succeeds where weak SINDy fails 194 times and the reverse never happens (exact sign test, $p = 8 \cdot 10^{-59}$). The gap is largest exactly where the grouping principle predicts it: on products

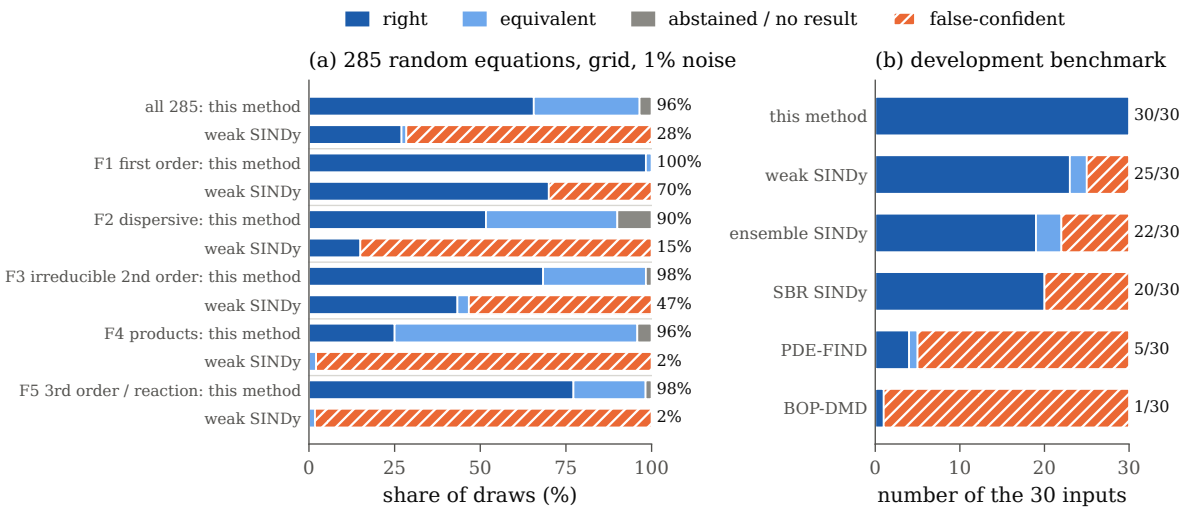


Figure 7. *Outcomes. (a) Large benchmark, overall and per family: share of draws that are right, equivalent, abstained or false-confident, with the right-or-equivalent share at the end of each bar. (b) Development benchmark:* 10 *equations* × 3 *designs. Hatched: a wrong equation returned without any warning.*

of factors (F4) and on higher-order and reaction factors (F5) weak SINDy is right on one draw in fifty, the method on 96–98%. Where both are right, the method's median relative coefficient error is 0.4%, against 1.4%. Counting every one of the 15 unfinished draws as a failure, the method is still right or equivalent on 275/300 = 91.7% (Wilson interval 88.0–94.3%); the unfinished draws are the slowest, not a random subset.

Development benchmark. Ten classical equations (transport, heat, advection–diffusion, Airy, Schrödinger, beam, wave, Klein–Gordon, telegraph and the running example), each in three designs: a 64×128 grid with noise 1%, the same grid with 5%, and 400 scattered points with 5%. The defaults were tuned on these equations, so this is not a held-out test; it compares the method with five baselines on classical cases (Figure 7(b), Table 3). The method recovers all 30 factor sets. The best baseline, weak SINDy, is right or equivalent on 25 and false-confident on 5, four of them products of factors. Repeating the two grid designs with three new fields and noise draws each (60 independent inputs), the method is right on all 60 with coefficient coverage 0.94 at nominal 0.90; weak SINDy is false-confident on 13.

Parameters and cost. The user-facing parameters are few: the starting and largest class ($C(3,4)$ and $C(4,8)$), the detection threshold $z = 3$, the goodness-of-fit level 10^{-3}, the evidence margin of one nat, and the number of posterior samples and bootstrap refits (8 and 16); section SM4 lists all of them. All were fixed during development and left unchanged for the large benchmark. The price of exploring a whole class of factored structures on every sample is time: a median of 127 s per input on the development benchmark and 13 minutes on the random equations, against seconds for library regression.

What these numbers do and do not show. All inputs are synthetic and the noise is white; the large benchmark uses one noise level, gridded data and one baseline. The claims we draw are therefore qualitative but sharp: on equations whose symbol is a product of factors, or

Table 3

Development benchmark: right or equivalent / false-confident, out of 10 *inputs per design, and the repetition with independent fields and noise (out of* 60*). Runtimes are wall-clock medians per input.*

Method	grid 1%	grid 5%	400 pts 5%	total	independent	median time
this method	10 / 0	10 / 0	10 / 0	30 / 0	60 / 0	127 s
weak SINDy	10 / 0	7 / 3	8 / 2	25 / 5	47 / 13	3 s
ensemble SINDy	9 / 1	7 / 3	6 / 4	22 / 8	–	5 s
SBR SINDy	8 / 2	6 / 4	6 / 4	20 / 10	–	14 s
PDE-FIND	1 / 9	1 / 9	3 / 7	5 / 25	–	0.6 s
BOP-DMD	1 / 9	0 / 10	0 / 10	1 / 29	–	1.5 s

Table 4

The kinds of claim made in the paper.

Kind	Statements
proved	groups are factors with overlap $\delta(P)$ (Theorem 2.3); terms are not groups (Proposition 2.5); stability via γ_N (Theorem 2.6) and its Bayesian form (Proposition 3.2); identification and its limits on finite data (Theorem 2.8 and Proposition 2.9); counting condition (Proposition 3.1)
design rule	cluster modes onto curves, not terms onto a library; labels instead of "identified"; decisions on the raw observations; growth of the class instead of coefficient penalties; minimum-norm split of the overlap
measured	275/285 right or equivalent and 0 confident wrong answers on random equations not used in development; 30/30 on the development benchmark and 60/60 on its independent repetitions; coefficient coverage 0.94 at nominal 0.90

has factors of higher order, grouping by factors recovers the structure where regression on a library of terms does not, and when the data do not suffice the labels and intervals warn instead of returning a wrong equation.

5. Discussion.

What is proved, what is designed, what is measured. Table 4 separates the three kinds of claim. The theorems hold under their hypotheses; the algorithm uses them as design rules, not as guarantees about its output; and the numbers are evidence on synthetic data.

Limits that no amount of data removes. Some limits are theorems, and the output is built to show them rather than hide them. *Error floor*: in the Gaussian model of subsection 3.4 every unbiased estimator of the components has variance at least σ_n^2/γ_N in the worst direction, so closely aligned groups are intrinsically hard to separate. *No data-free threshold*: whether a numerical symbol "factors" depends on the noise level, which is why decisions are made by evidence and not by a tolerance on coefficients. *Repeated factors*: finite noisy data cannot distinguish P_i^2 from two nearby factors, hence the label UNDECIDED. *No certificate*: by Proposition 2.9, outputs are compatibility statements with uncertainty, never identifications.

Beyond the linear ring. The two groupings do not depend on the equation being linear with constant coefficients in one space dimension, and there are concrete reasons to expect the algorithm to extend. The matrix pencil reads the temporal rates at every wavenumber

vector $\kappa \in \mathbb{R}^d$ exactly as in one dimension, and the groups become hypersurfaces $Q_j(\lambda, i\kappa) = 0$ that grouping I can fit in the same way. For slowly varying coefficients a windowed transform gives local rates $\lambda(\kappa; x_0)$, and each group becomes a family of curves indexed by position. For a nonlinear equation, a short-time expansion shows that the flow is its linearization plus a quadratic interaction term, so modes of the linearized flow can still be grouped on short windows. What does not carry over unchanged is the theory: in higher dimension two factors can share a whole curve of zeros, the operators with variable coefficients form a non-commutative algebra without unique factorization, and nonlinear solutions do not superpose. The first task in each direction is therefore the one carried out here for the linear case: decide which pieces of the equation a solution carries, how uniquely and how stably, and only then design the learning procedure around them (section SM6).

6. Conclusion. Groups are factors. This one statement, made precise by Theorem 2.3 and its companions, determines what an equation-learning algorithm should look for (modes on the curves of the factors), how it should combine evidence (clustering factors across posterior samples), how it should measure uncertainty (through the Friedrichs angle), and what it may claim (compatibility, never certification). On 285 random equations never used in its development the resulting method is right or equivalent on 96.5% and never wrong without warning, while regression on a library of terms is wrong without warning on most products and higher-order factors.

Code and data availability. The code, the benchmark scripts and every stored result behind the numbers of this paper are at https://github.com/FlavioApric/pde-factor-fit [14]; a snapshot is included in the supplementary materials.

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